

# Package ‘GTAPViz’

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**Title** R Package for Automating 'GTAP' Data Processing and Visualization

**Version** 1.1.0

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**Description** An R package designed to streamline the processing and visualization of 'GTAP' model results. While primarily built for 'GTAP' simulations, it also supports other .har and .sl4 datasets. As an extension of HARplus, GTAPViz simplifies 'GTAP' data extraction and visualization, reducing manual effort and enhancing efficiency. With intuitive commands and automated adjustments, users can create, customize, and export high-quality visualizations for economic analysis and research reporting with minimal coding.

**License** MIT + file LICENSE

**Encoding** UTF-8

**RoxygenNote** 7.3.2

**BugReports** <https://github.com/bodysbobb/GTAPViz/issues>

**URL** <https://bodysbobb.github.io/GTAPViz/>

**Imports** HARplus,

ggplot2,  
dplyr,  
tidyr,  
openxlsx,  
colorspace,  
grDevices,  
scales,  
utils,  
methods,  
stringdist,  
stats,  
glue,  
openxlsx2

**VignetteBuilder** knitr

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rmarkdown,  
devtools

**Depends** R (>= 3.5)

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|                  |                                             |
|------------------|---------------------------------------------|
| add_mapping_info | <i>Add Mapping Information to GTAP Data</i> |
|------------------|---------------------------------------------|

---

Description

Adds descriptions and unit information to GTAP data based on a specified mapping mode. Supports external mappings or default GTAPv7 mappings, allowing users to enrich datasets with standardized metadata.

Usage

```
add_mapping_info(  
  data_list,  
  external_map = NULL,  
  mapping = "GTAPv7",  
  description_info = TRUE,  
  unit_info = TRUE  
)
```

Arguments

|              |                                                                                                                                                                                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data_list    | A data structure containing GTAP variables.                                                                                                                                                                                                         |
| external_map | Optional data frame with mapping information (must include "Variable", "Description", and "Unit" columns).                                                                                                                                          |
| mapping      | Character. Controls how descriptions and units are added: "GTAPv7" (default) uses standard GTAPv7 definitions; "Yes" uses only the supplied 'external_map'; "No" skips mapping; "Mix" prioritizes 'external_map', falling back to GTAPv7 if needed. |

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| description_info | Logical. If 'TRUE', adds description information to the data. |
| unit_info        | Logical. If 'TRUE', adds unit information to the data.        |

### Details

The 'mapping' argument controls how description and unit information are added to the dataset:

- "GTAPv7" (default): Applies standardized GTAPv7 metadata for variable descriptions and units.
- "Yes": Uses only the descriptions and units from the 'external\_map' provided by the user. No fallback to GTAPv7.
- "No": Skips mapping entirely. No descriptions or units are added.
- "Mix": Prioritizes entries in 'external\_map', but fills in any missing values using GTAPv7 meta-data where available.

### Value

A data structure with added mapping information, preserving the original structure.

### Author(s)

Pattawee Puangchit

### See Also

[convert\\_units](#), [rename\\_value](#)

### Examples

```
# Load Sample Data:
sl4_data1 <- HARplus::load_sl4x(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPViz"))

# Get Data by Variable Name
sl4_data1 <- HARplus::get_data_by_var(c("qgdp", "EV"), sl4_data1)

# Add mapping using GTAPv7 defaults
gtap_data <- add_mapping_info(sl4_data1, mapping = "GTAPv7")

# Use an external mapping file
my_mapping <- data.frame(Variable = c("qgdp", "EV"),
                        Description = c("Real GDP", "Welfare"),
                        Unit = c("percent", "millionUSD")) # <- Fixed closing quote

gtap_data <- add_mapping_info(sl4_data1, external_map = my_mapping,
                             mapping = "Mix")
```

### Description

Processes GTAP data from SL4 and HAR files with options for exporting and preparing plot-ready data.

## Usage

```

auto_gtap_data(
  experiment,
  project_path = NULL,
  input_path = NULL,
  output_path = NULL,
  sl4_suffix = "",
  har_suffix = "-WEL",
  mapping_info = "GTAPv7",
  process_sl4_vars = NULL,
  process_har_vars = NULL,
  sl4_mapping_info = NULL,
  har_mapping_info = NULL,
  sl4_extract_method = "group_data_by_dims",
  har_extract_method = "get_data_by_var",
  sl4_priority = NULL,
  har_priority = NULL,
  sl4_convert_unit = NULL,
  har_convert_unit = NULL,
  decimals = 4,
  rename_columns = TRUE,
  region_select = NULL,
  sector_select = NULL,
  subtotal_level = FALSE,
  plot_data = FALSE,
  output_formats = NULL,
  sl4_output_name = "sl4.plot.data",
  har_output_name = "har.plot.data",
  macro_output_name = "GTAPMacro",
  add_scenario_ranking = FALSE,
  rank_column = "ScenarioRank"
)

```

## Arguments

|                               |                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>experiment</code>       | Character vector. Case names to process.                                                                    |
| <code>project_path</code>     | Character. Path to project folder with "in" and "out" subfolders.                                           |
| <code>input_path</code>       | Character. Input folder path. Overrides 'project_path/in' if specified.                                     |
| <code>output_path</code>      | Character. Output folder path. Overrides 'project_path/out' if specified.                                   |
| <code>sl4_suffix</code>       | Character. Custom suffix for SL4 files (e.g., "", "-custom").                                               |
| <code>har_suffix</code>       | Character. Custom suffix for HAR files (e.g., "-WEL").                                                      |
| <code>mapping_info</code>     | Character. Metadata mode: "GTAPv7" (default), "Yes", "No", or "Mix".                                        |
| <code>process_sl4_vars</code> | Data frame, NULL, or FALSE. Variables to extract from SL4: NULL = extract all; FALSE = skip SL4 processing. |
| <code>process_har_vars</code> | Data frame, NULL, or FALSE. Variables to extract from HAR: NULL = extract all; FALSE = skip HAR processing. |
| <code>sl4_mapping_info</code> | Data frame or NULL. Mapping info for SL4 (must include "Variable", "Description", and "Unit").              |

|                      |                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------|
| har_mapping_info     | Data frame or NULL. Mapping info for HAR (same structure as SL4).                                                 |
| sl4_extract_method   | Character. Method for SL4 extraction: "get_data_by_dims", "get_data_by_var", or "group_data_by_dims".             |
| har_extract_method   | Character. Method for HAR extraction: same options as above.                                                      |
| sl4_priority         | Optional list. Priority rules for SL4 data grouping.                                                              |
| har_priority         | Optional list. Priority rules for HAR data grouping.                                                              |
| sl4_convert_unit     | Character. Optional SL4 unit conversion: "mil2bil", "bil2mil", "pct2frac", or "frac2pct". Default is NULL.        |
| har_convert_unit     | Character. Optional HAR unit conversion: same options as above. Default is NULL.                                  |
| decimals             | Integer. Number of decimal places to round numeric values to. Default is 2. Set to NULL to disable rounding.      |
| rename_columns       | Logical. If TRUE (default), renames GTAP column identifiers to more readable format                               |
| region_select        | Optional character vector. Regions to include.                                                                    |
| sector_select        | Optional character vector. Sectors to include.                                                                    |
| subtotal_level       | Logical. Whether to include subtotal rows. Default is FALSE.                                                      |
| plot_data            | Logical. If TRUE, prepares and assigns plot-ready data.                                                           |
| output_formats       | Character vector or list. Output formats to export (valid: "csv", "stata", "rds", "txt").                         |
| sl4_output_name      | Character. Output variable name for SL4 results. Default is "sl4.plot.data".                                      |
| har_output_name      | Character. Output variable name for HAR results. Default is "har.plot.data".                                      |
| macro_output_name    | Character. Output variable name for macro results. Default is "GTAPMacro".                                        |
| add_scenario_ranking | Logical or "merged". If TRUE, adds ranking column; if "merged", also prefixes experiment names. Default is FALSE. |
| rank_column          | Character. Name of the ranking column to add. Default is "ScenarioRank".                                          |

## Details

This function automates the workflow for processing GTAP model outputs, with flexible output options and optional filtering for region- and sector-specific data.

The key parameters 'process\_sl4\_vars' and 'process\_har\_vars' accept three different input types:

- **Data frame:** Contains variable mappings with required "Variable" column. When provided, only specified variables will be extracted.
- **NULL:** Extracts all available variables from the respective file type.
- **FALSE:** Completely skips processing of that file type, allowing the function to focus only on the other file type.

The ‘mapping\_info’ parameter controls how descriptions and units are assigned:

- **GTAPv7**: Uses standard GTAPv7 definitions (default).
- **Yes**: Uses only the supplied descriptions and units from ‘sl4\_mapping\_info’ / ‘har\_mapping\_info’.
- **No**: Does not add any descriptions or units.
- **Mix**: Prioritizes supplied descriptions and units, falling back to GTAPv7 for any missing values.

### Value

A processed dataset, with options for exporting and visualization.

### Author(s)

Pattawee Puangchit

### See Also

[add\\_mapping\\_info](#), [gtap\\_macros\\_data](#)

### Examples

```
# Input Path:
input_path <- system.file("extdata/in", package = "GTAPViz")

# GTAP Macro Variables from 2 .sl4 Files named (EXP1, EXP2)
# Note: No need to add .sl4 to the experiment name
gtap_data <- auto_gtap_data(experiment = c("EXP1", "EXP2"),
                           input_path = input_path, subtotal_level = FALSE,
                           process_sl4_vars = NULL, process_har_vars = NULL,
                           mapping_info = "GTAPv7", plot_data = TRUE)
```

---

comparison\_plot

*Create Comparative Bar Charts from HAR and SL4 Data*

---

### Description

Generates comparative bar charts using GTAP data, allowing multiple visualization options such as panel facets, split grouping, and customizable styles.

### Usage

```
comparison_plot(
  data,
  filter_var = NULL,
  x_axis_from,
  split_by = NULL,
  panel_var = "Experiment",
  variable_col = "Variable",
  unit_col = "Unit",
  desc_col = "Description",
```

```

invert_pane = FALSE,
separate_figure = FALSE,
var_name_by_description = FALSE,
add_var_info = FALSE,
output_path = NULL,
export_picture = TRUE,
export_as_pdf = FALSE,
export_config = NULL,
plot_style_config = NULL
)

```

## Arguments

|                                      |                                                                                                                                                                                                                                                                                    |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>data</code>                    | A data frame or a list of data frames containing GTAP results.                                                                                                                                                                                                                     |
| <code>filter_var</code>              | Vector or data frame. If a vector, filters the values in <code>'x_axis_from'</code> . If a data frame, filters <code>'value_col'</code> based on matching <code>'variable_col'</code> values.                                                                                      |
| <code>x_axis_from</code>             | Character. Column name for x-axis categories (e.g., "REG", "Sector").                                                                                                                                                                                                              |
| <code>split_by</code>                | Character or vector. Column name(s) to generate separate plots for each unique value (e.g., "COMM", "REG", "Variable"). NULL creates a single aggregated plot, appropriate for macro-level analysis.                                                                               |
| <code>panel_var</code>               | Character. Column for panel facets (default: "Experiment").                                                                                                                                                                                                                        |
| <code>variable_col</code>            | Character. Column containing variable identifiers (default: "Variable").                                                                                                                                                                                                           |
| <code>unit_col</code>                | Character. Column containing unit information (default: "Unit").                                                                                                                                                                                                                   |
| <code>desc_col</code>                | Character. Column containing variable descriptions (default: "Description").                                                                                                                                                                                                       |
| <code>invert_pane</code>             | Logical. If TRUE, creates horizontal bars instead of vertical ones (default: FALSE).                                                                                                                                                                                               |
| <code>separate_figure</code>         | Logical. If TRUE, generates separate figures per panel value (default: FALSE).                                                                                                                                                                                                     |
| <code>var_name_by_description</code> | Logical. If TRUE, uses descriptions instead of variable codes in titles (default: FALSE).                                                                                                                                                                                          |
| <code>add_var_info</code>            | Logical. If TRUE, adds the variable code in parentheses after the description (default: FALSE).                                                                                                                                                                                    |
| <code>output_path</code>             | Character. Directory path for saving output files. If NULL, plots are only returned in R.                                                                                                                                                                                          |
| <code>export_picture</code>          | Logical. If TRUE, exports plots as image files (default: TRUE).                                                                                                                                                                                                                    |
| <code>export_as_pdf</code>           | Logical or "merged". If TRUE, exports separate PDFs; if "merged", creates a multi-page PDF; if FALSE, skips PDF (default: FALSE).                                                                                                                                                  |
| <code>export_config</code>           | List. Export settings, including: <ul style="list-style-type: none"> <li>• <code>'width'</code>: Output file width (in inches).</li> <li>• <code>'height'</code>: Output file height (in inches).</li> <li>• Additional settings—see <code>'?get_export_config'</code>.</li> </ul> |
| <code>plot_style_config</code>       | List. Custom plot styles—see <code>'?get_plot_style_config'</code> .                                                                                                                                                                                                               |

## Value

A ggplot2 object for a single plot or a list of ggplot2 objects for multiple plots.

**Author(s)**

Pattawee Puangchit

**See Also**

[get\\_plot\\_style\\_config](#), [get\\_export\\_config](#), [detail\\_plot](#), [stack\\_plot](#)

**Examples**

```
# Input Path:
input_path <- system.file("extdata/in", package = "GTAPViz")

# GTAP Macro Variables from 2 .sl4 Files named (EXP1, EXP2)
# Note: No need to add .sl4 to the experiment name
gtap_data <- auto_gtap_data(experiment = c("EXP1", "EXP2"),
                           input_path = input_path, subtotal_level = FALSE,
                           process_sl4_vars = NULL, process_har_vars = NULL,
                           mapping_info = "GTAPv7", plot_data = TRUE)

# Basic usage with data frame
p1 <- comparison_plot(
  data = sl4.plot.data[["1D"]][["Region"]],
  x_axis_from = "Region",
  panel_var = "Experiment",
  filter_var = c("qgdp", "EV"),
  output_path = tempdir(),
  export_picture = FALSE,
  export_as_pdf = FALSE,
)

# Split by commodity with custom styling and export options
p2 <- comparison_plot(
  data = sl4.plot.data[["1D"]][["Region"]],
  x_axis_from = "Region",
  split_by = "Variable",
  panel_var = "Experiment",
  filter_var = c("qgdp", "EV"),
  var_name_by_description = TRUE,
  output_path = tempdir(),
  export_picture = FALSE,
  export_as_pdf = FALSE,
  export_config = list(
    file_name = "commodity_impacts",
    width = 12,
    height = 8
  ),
  plot_style_config = list(
    color_tone = "economic",
    title_size = 16,
    show_grid_major_y = TRUE
  )
)
```



convert\_units

*Convert Units in GTAP Data***Description**

Converts values in a dataset to different units based on predefined transformations or custom scaling. Supports manual and automatic conversions for economic and trade-related metrics.

**Usage**

```
convert_units(
  data,
  change_unit_from = NULL,
  change_unit_to = NULL,
  adjustment = NULL,
  value_col = "Value",
  unit_col = "Unit",
  variable_select = NULL,
  variable_col = "Variable",
  scale_auto = NULL
)
```

**Arguments**

|                  |                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data             | A data structure (list, data frame, or nested combination).                                                                                                                                                                                                                                                                                                                               |
| change_unit_from | Character vector. Units to be converted (case-insensitive).                                                                                                                                                                                                                                                                                                                               |
| change_unit_to   | Character vector. Target units corresponding to 'change_unit_from'.                                                                                                                                                                                                                                                                                                                       |
| adjustment       | Character or numeric vector. Specifies conversion operations (e.g., <code>"/1000"</code> to convert million to billion).                                                                                                                                                                                                                                                                  |
| value_col        | Character. Column name containing values to adjust (default: <code>"Value"</code> ).                                                                                                                                                                                                                                                                                                      |
| unit_col         | Character. Column name containing unit information (default: <code>"Unit"</code> ).                                                                                                                                                                                                                                                                                                       |
| variable_select  | Optional character vector. If provided, only these variables are converted.                                                                                                                                                                                                                                                                                                               |
| variable_col     | Character. Column name containing variable identifiers (default: <code>"Variable"</code> ).                                                                                                                                                                                                                                                                                               |
| scale_auto       | Optional character vector of predefined conversion rules: - <code>"mil2bil"</code> : Converts million USD to billion USD (divides by 1000). - <code>"bil2mil"</code> : Converts billion USD to million USD (multiplies by 1000). - <code>"pct2frac"</code> : Converts percent to fraction (divides by 100). - <code>"frac2pct"</code> : Converts fraction to percent (multiplies by 100). |

**Details**

If both `'change_unit_from'` and `'scale_auto'` are provided, the function prompts the user to choose between manual and automatic conversion.

**Value**

A data structure with values converted to the specified units.

**Author(s)**

Pattawee Puangchit

**See Also**[add\\_mapping\\_info](#), [rename\\_value](#), [sort\\_plot\\_data](#)**Examples**

```
# Load Sample Data:
sl4_data1 <- HARplus::load_sl4x(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPViz"))

# Get Data by Variable Name
sl4_data1 <- HARplus::get_data_by_var(c("qgdp", "EV"), sl4_data1)

# Convert million USD to billion USD
gtap_data <- convert_units(sl4_data1,
  change_unit_from = "million USD",
  change_unit_to = "billion USD",
  adjustment = "/1000"
)

# Automatic conversion from percent to fraction
gtap_data <- convert_units(sl4_data1, scale_auto = "pct2frac")
```

---

create\_export\_config    *Create an Export Configuration*

---

**Description**

Creates a configuration list for controlling plot export settings. This function provides auto-completion for export options.

**Usage**

```
create_export_config(
  file_name = "gtap_plots",
  width = NULL,
  height = NULL,
  dpi = 300,
  bg = "white",
  limitsize = FALSE
)
```

**Arguments**

|           |                                                                       |
|-----------|-----------------------------------------------------------------------|
| file_name | Character. Base name for exported files. Default: "gtap_plots".       |
| width     | Numeric. Width of output in inches. Default: NULL (auto-calculated).  |
| height    | Numeric. Height of output in inches. Default: NULL (auto-calculated). |
| dpi       | Numeric. Resolution for PNG export. Default: 300.                     |

bg                    Character. Background color. Default: "white".  
 limitsize           Logical. Whether to limit size. Default: FALSE.

### Value

A list with export configuration parameters.

### Author(s)

Pattawee Puangchit

### Examples

```
# Default export configuration
default_export <- create_export_config()

# Custom export configuration
custom_export <- create_export_config(
  file_name = "regional_impacts",
  width = 12,
  height = 8,
  dpi = 600
)
```

---

|                   |                                          |
|-------------------|------------------------------------------|
| create_plot_style | <i>Create a Plot Style Configuration</i> |
|-------------------|------------------------------------------|

---

### Description

Creates a configuration list for plot styling that can be used with GTAPViz plotting functions. This function provides auto-completion for style options while maintaining compatibility with direct list specification.

### Usage

```
create_plot_style(
  show_title = TRUE,
  title_face = "bold",
  title_size = 20,
  title_hjust = 0.5,
  add_unit_to_title = TRUE,
  title_margin = c(10, 0, 10, 0),
  title_format = list(type = "standard", text = "", sep = ""),
  show_x_axis_title = TRUE,
  x_axis_title_face = "bold",
  x_axis_title_size = 16,
  x_axis_title_margin = c(25, 25, 0, 0),
  show_x_axis_labels = TRUE,
  x_axis_text_face = "plain",
  x_axis_text_size = 14,
  x_axis_text_angle = 0,
  x_axis_text_hjust = 0,
```

```

x_axis_description = "",
show_y_axis_title = TRUE,
y_axis_title_face = "bold",
y_axis_title_size = 16,
y_axis_title_margin = c(25, 25, 0, 0),
show_y_axis_labels = TRUE,
y_axis_text_face = "plain",
y_axis_text_size = 14,
y_axis_text_angle = 0,
y_axis_text_hjust = 0,
y_axis_description = "",
show_axis_titles_on_all_facets = TRUE,
show_value_labels = TRUE,
value_label_face = "plain",
value_label_size = 5,
value_label_position = "above",
value_label_decimal_places = 2,
show_legend = FALSE,
show_legend_title = FALSE,
legend_position = "bottom",
legend_title_face = "bold",
legend_text_face = "plain",
legend_text_size = 14,
strip_face = "bold",
strip_text_size = 16,
strip_background = "lightgrey",
strip_text_margin = c(10, 0, 10, 0),
panel_spacing = 2,
panel_rows = NULL,
panel_cols = NULL,
theme = NULL,
color_tone = NULL,
color_palette_type = "qualitative",
positive_color = "#2E8B57",
negative_color = "#CD5C5C",
background_color = "white",
grid_color = "grey90",
show_grid_major_x = FALSE,
show_grid_major_y = FALSE,
show_grid_minor_x = FALSE,
show_grid_minor_y = FALSE,
show_zero_line = TRUE,
zero_line_type = "dashed",
zero_line_color = "black",
zero_line_size = 0.5,
zero_line_position = 0,
bar_width = 0.9,
bar_spacing = 0.9,
scale_limit = NULL,
scale_increment = NULL,
expansion_y_mult = c(0.05, 0.1),
expansion_x_mult = c(0.05, 0.05),

```

```

    all_font_size = 1,
    sort_data_by_value = FALSE,
    plot.margin = c(10, 25, 10, 10)
)

```

## Arguments

|                                  |                                                                                                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>show_title</code>          | Logical. Show or hide the plot title. Default: TRUE                                                                                                               |
| <code>title_face</code>          | Character. Font face ("bold", "plain", "italic"). Default: "bold"                                                                                                 |
| <code>title_size</code>          | Numeric. Font size of title. Default: 20                                                                                                                          |
| <code>title_hjust</code>         | Numeric. Horizontal alignment (0 = left, 1 = right). Default: 0.5                                                                                                 |
| <code>add_unit_to_title</code>   | Logical. Append unit to title if applicable. Default: TRUE                                                                                                        |
| <code>title_margin</code>        | Numeric vector c(top, right, bottom, left). Default: c(10, 0, 10, 0)                                                                                              |
| <code>title_format</code>        | List or function output. Title formatting options. Can be created with <code>create_title_format()</code> . Default: list(type = "standard", text = "", sep = "") |
| <code>show_x_axis_title</code>   | Logical. Show or hide x-axis title. Default: TRUE                                                                                                                 |
| <code>x_axis_title_face</code>   | Character. Font face for x-axis title. Default: "bold"                                                                                                            |
| <code>x_axis_title_size</code>   | Numeric. Font size of x-axis title. Default: 16                                                                                                                   |
| <code>x_axis_title_margin</code> | Numeric vector c(top, right, bottom, left). Default: c(25, 25, 0, 0)                                                                                              |
| <code>show_x_axis_labels</code>  | Logical. Show or hide x-axis labels. Default: TRUE                                                                                                                |
| <code>x_axis_text_face</code>    | Character. Font face for x-axis labels. Default: "plain"                                                                                                          |
| <code>x_axis_text_size</code>    | Numeric. Font size of x-axis labels. Default: 14                                                                                                                  |
| <code>x_axis_text_angle</code>   | Numeric. Angle of x-axis labels. Default: 0                                                                                                                       |
| <code>x_axis_text_hjust</code>   | Numeric. Horizontal justification of x-axis labels. Default: 0                                                                                                    |
| <code>x_axis_description</code>  | Character. Optional description for the x-axis. Default: ""                                                                                                       |
| <code>show_y_axis_title</code>   | Logical. Show or hide y-axis title. Default: TRUE                                                                                                                 |
| <code>y_axis_title_face</code>   | Character. Font face for y-axis title. Default: "bold"                                                                                                            |
| <code>y_axis_title_size</code>   | Numeric. Font size of y-axis title. Default: 16                                                                                                                   |
| <code>y_axis_title_margin</code> | Numeric vector c(top, right, bottom, left). Default: c(25, 25, 0, 0)                                                                                              |
| <code>show_y_axis_labels</code>  | Logical. Show or hide y-axis labels. Default: TRUE                                                                                                                |
| <code>y_axis_text_face</code>    | Character. Font face for y-axis labels. Default: "plain"                                                                                                          |

**y\_axis\_text\_size**      Numeric. Font size of y-axis labels. Default: 14  
**y\_axis\_text\_angle**      Numeric. Angle of y-axis labels. Default: 0  
**y\_axis\_text\_hjust**      Numeric. Horizontal justification of y-axis labels. Default: 0  
**y\_axis\_description**      Character. Optional description for the y-axis. Default: ""  
**show\_axis\_titles\_on\_all\_facets**      Logical. Show axis titles on all facets. Default: TRUE  
**show\_value\_labels**      Logical. Show or hide value labels. Default: TRUE  
**value\_label\_face**      Character. Font face for value labels. Default: "plain"  
**value\_label\_size**      Numeric. Font size of value labels. Default: 5  
**value\_label\_position**      Character. Position of value labels ("above", "outside", "top"). Default: "above"  
**value\_label\_decimal\_places**      Numeric. Number of decimal places in value labels. Default: 2  
**show\_legend**      Logical. Show or hide legend. Default: FALSE  
**show\_legend\_title**      Logical. Show or hide legend title. Default: FALSE  
**legend\_position**      Character. Legend position ("none", "bottom", "right"). Default: "bottom"  
**legend\_title\_face**      Character. Font face for legend title. Default: "bold"  
**legend\_text\_face**      Character. Font face for legend text. Default: "plain"  
**legend\_text\_size**      Numeric. Font size of legend text. Default: 14  
**strip\_face**      Character. Font face for panel strip. Default: "bold"  
**strip\_text\_size**      Numeric. Font size for panel strip. Default: 16  
**strip\_background**      Character. Background color of strip. Default: "lightgrey"  
**strip\_text\_margin**      Numeric vector c(top, right, bottom, left). Default: c(10, 0, 10, 0)  
**panel\_spacing**      Numeric. Spacing between panels. Default: 2  
**panel\_rows**      Numeric or NULL. Number of rows in panel layout. Default: NULL  
**panel\_cols**      Numeric or NULL. Number of columns in panel layout. Default: NULL  
**theme**      ggplot2 theme or NULL. Custom ggplot theme. Default: NULL  
**color\_tone**      Character or NULL. Base color theme. Default: NULL  
**color\_palette\_type**      Character. Type of color palette ('qualitative', 'sequential', 'diverging'). Default: "qualitative"  
**positive\_color**      Character. Color for positive values. Default: "#2E8B57"

|                    |                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------|
| negative_color     | Character. Color for negative values. Default: "#CD5C5C"                                               |
| background_color   | Character. Background color of plot. Default: "white"                                                  |
| grid_color         | Character. Color of grid lines. Default: "grey90"                                                      |
| show_grid_major_x  | Logical. Show major grid lines on x-axis. Default: FALSE                                               |
| show_grid_major_y  | Logical. Show major grid lines on y-axis. Default: FALSE                                               |
| show_grid_minor_x  | Logical. Show minor grid lines on x-axis. Default: FALSE                                               |
| show_grid_minor_y  | Logical. Show minor grid lines on y-axis. Default: FALSE                                               |
| show_zero_line     | Logical. Show or hide zero line. Default: TRUE                                                         |
| zero_line_type     | Character. Line type ("solid", "dashed", "dotted"). Default: "dashed"                                  |
| zero_line_color    | Character. Color of zero line. Default: "black"                                                        |
| zero_line_size     | Numeric. Line thickness of zero line. Default: 0.5                                                     |
| zero_line_position | Numeric. Position of the zero line. Default: 0                                                         |
| bar_width          | Numeric. Width of bars. Default: 0.9                                                                   |
| bar_spacing        | Numeric. Spacing between groups of bars. Default: 0.9                                                  |
| scale_limit        | Numeric vector of length 2 or NULL. Manual limits for value axis. Default: NULL                        |
| scale_increment    | Numeric or NULL. Step size for axis tick marks. Default: NULL                                          |
| expansion_y_mult   | Numeric vector. Y-axis expansion. Default: c(0.05, 0.1)                                                |
| expansion_x_mult   | Numeric vector. X-axis expansion. Default: c(0.05, 0.05)                                               |
| all_font_size      | Numeric. Master control for all font sizes. Default: 1                                                 |
| sort_data_by_value | Logical. Whether to sort data by value. Default: FALSE                                                 |
| plot.margin        | Numeric vector c(top, right, bottom, left). Margins around the entire plot. Default: c(10, 25, 10, 10) |

**Value**

A list containing all style configuration parameters

**Author(s)**

Pattawee Puangchit

**Examples**

```
# Create default style
default_style <- create_plot_style()

# Create customized style with title formatting
custom_style <- create_plot_style(
  color_tone = "gtap",
  title_size = 24,
  title_format = create_title_format(
    type = "prefix",
    text = "Impact on",
    sep = " "
  ),
  bar_width = 0.5,
  x_axis_text_angle = 45
)
```

---

|                     |                                     |
|---------------------|-------------------------------------|
| create_title_format | Create a Title Format Configuration |
|---------------------|-------------------------------------|

---

**Description**

Creates a configuration list for controlling plot title formatting. This function provides auto-completion for title format options.

**Usage**

```
create_title_format(type = "standard", text = "", sep = "")
```

**Arguments**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type | Character. The type of title format. Options: <ul style="list-style-type: none"> <li>"standard" - Default automatic title based on variable, description, and unit.</li> <li>"prefix" - Adds text before the automatic title.</li> <li>"suffix" - Adds text after the automatic title.</li> <li>"full" - Uses only the specified text as the title.</li> <li>"dynamic" - Creates a custom title using column values.</li> </ul> |
| text | Character. The text to add as prefix, suffix, full title, or column template for dynamic titles.                                                                                                                                                                                                                                                                                                                                |
| sep  | Character. Separator between prefix/suffix and main title. Default: "".                                                                                                                                                                                                                                                                                                                                                         |

**Value**

A list with title format configuration parameters.

**Author(s)**

Pattawee Puangchit



## Examples

```
# Standard auto-generated title
standard_title <- create_title_format()

# Prefix title
prefix_title <- create_title_format(
  type = "prefix",
  text = "Impact on",
  sep = " "
)

# Dynamic title using column values
dynamic_title <- create_title_format(
  type = "dynamic",
  text = "Impact on {Variable} in {Region}"
)
```

---

detail\_plot

---

Create Comprehensive Bar Charts from HAR and SL4 Data

---

## Description

Generates detailed bar charts to visualize the distribution of impacts across multiple dimensions. The function supports top impact filtering, color coding, and flexible visualization settings.

## Usage

```
detail_plot(
  data,
  filter_var = NULL,
  x_axis_from,
  split_by = NULL,
  panel_var = "Experiment",
  variable_col = "Variable",
  unit_col = "Unit",
  desc_col = "Description",
  var_name_by_description = FALSE,
  add_var_info = FALSE,
  top_impact = NULL,
  invert_pane = FALSE,
  separate_figure = FALSE,
  output_path = NULL,
  export_picture = TRUE,
  export_as_pdf = FALSE,
  export_config = NULL,
  plot_style_config = NULL
)
```

## Arguments

**data** A data frame or a list of data frames containing GTAP results.

|                                      |                                                                                                                                                                                                                                                                                    |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>filter_var</code>              | Vector or data frame. If a vector, filters the values in <code>'x_axis_from'</code> . If a data frame, filters <code>'value_col'</code> based on matching <code>'variable_col'</code> values.                                                                                      |
| <code>x_axis_from</code>             | Character. Column name for x-axis categories (e.g., "REG", "Sector").                                                                                                                                                                                                              |
| <code>split_by</code>                | Character or vector. Column name(s) to generate separate plots for each unique value (e.g., "COMM", "REG", "Variable"). NULL creates a single aggregated plot, appropriate for macro-level analysis.                                                                               |
| <code>panel_var</code>               | Character. Column for panel facets (default: "Experiment").                                                                                                                                                                                                                        |
| <code>variable_col</code>            | Character. Column containing variable identifiers (default: "Variable").                                                                                                                                                                                                           |
| <code>unit_col</code>                | Character. Column containing unit information (default: "Unit").                                                                                                                                                                                                                   |
| <code>desc_col</code>                | Character. Column containing variable descriptions (default: "Description").                                                                                                                                                                                                       |
| <code>var_name_by_description</code> | Logical. If TRUE, uses descriptions instead of variable codes in titles (default: FALSE).                                                                                                                                                                                          |
| <code>add_var_info</code>            | Logical. If TRUE, adds the variable code in parentheses after the description (default: FALSE).                                                                                                                                                                                    |
| <code>top_impact</code>              | Numeric or NULL. If specified, shows only the top N impactful values; NULL shows all values.                                                                                                                                                                                       |
| <code>invert_pane</code>             | Logical. If TRUE, creates horizontal bars instead of vertical ones (default: FALSE).                                                                                                                                                                                               |
| <code>separate_figure</code>         | Logical. If TRUE, generates separate figures per panel value (default: FALSE).                                                                                                                                                                                                     |
| <code>output_path</code>             | Character. Directory path for saving output files. If NULL, plots are only returned in R.                                                                                                                                                                                          |
| <code>export_picture</code>          | Logical. If TRUE, exports plots as image files (default: TRUE).                                                                                                                                                                                                                    |
| <code>export_as_pdf</code>           | Logical or "merged". If TRUE, exports separate PDFs; if "merged", creates a multi-page PDF; if FALSE, skips PDF (default: FALSE).                                                                                                                                                  |
| <code>export_config</code>           | List. Export settings, including: <ul style="list-style-type: none"> <li>• <code>'width'</code>: Output file width (in inches).</li> <li>• <code>'height'</code>: Output file height (in inches).</li> <li>• Additional settings—see <code>'?get_export_config'</code>.</li> </ul> |
| <code>plot_style_config</code>       | List. Custom plot styles—see <code>'?get_plot_style_config'</code> .                                                                                                                                                                                                               |

**Value**

A ggplot2 object for a single plot or a list of ggplot2 objects for multiple plots.

**Author(s)**

Pattawee Puangchit

**See Also**

[get\\_plot\\_style\\_config](#), [get\\_export\\_config](#), [comparison\\_plot](#), [stack\\_plot](#)

## Examples

```
# Input Path:
input_path <- system.file("extdata/in", package = "GTAPViz")

# GTAP Macro Variables from 2 .sl4 Files named (EXP1, EXP2)
# Note: No need to add .sl4 to the experiment name
gtap_data <- auto_gtap_data(experiment = c("EXP1", "EXP2"),
                           input_path = input_path, subtotal_level = FALSE,
                           process_sl4_vars = NULL, process_har_vars = NULL,
                           mapping_info = "GTAPv7", plot_data = TRUE)

# Basic usage with data frame
detail_plot(sl4.plot.data[["2D"]],
            x_axis_from = "Sector",
            split_by = "Region",
            filter_var = "qo",

            top_impact = NULL,
            var_name_by_description = TRUE,

            invert_pane = TRUE,
            separate_figure = FALSE,

            export_config = list(
              width = 45,
              height = 20
            ),

            export_picture = FALSE,
            export_as_pdf = FALSE,
            output_path = tempdir(),

            plot_style_config = list(
              positive_color = "#2E8B57",
              negative_color = "#CD5C5C",
              panel_rows = 1,
              panel_cols = NULL,
              show_axis_titles_on_all_facets = FALSE,
              y_axis_text_size = 25,
              bar_width = 0.6,
              all_font_size = 1.1
            ))
```

---

get\_all\_config

*Get Plot and Export Configurations*


---

## Description

Returns a list containing plot style configuration and export configuration. The function supports retrieving a single plot style or all available styles.

**Usage**

```
get_all_config(
  plot_style = "default",
  config = NULL,
  export_config = NULL,
  printing = FALSE
)
```

**Arguments**

|               |                                                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| plot_style    | Character. The plot style to retrieve: "comparison", "detail", "stack", or "all". Default is "all", which returns configurations for all styles. |
| config        | List. Optional custom style configuration to override defaults.                                                                                  |
| export_config | List. Optional custom export configuration to override defaults.                                                                                 |
| printing      | Logical. If TRUE, prints a formatted code snippet that can be copied and pasted to recreate the configuration. Default is FALSE.                 |

**Value**

A list with two components:

- plot\_style\_config: Plot style configuration(s). If plot\_style="all", contains a nested list with configurations for all styles.
- export\_config: Export configuration as a data frame.

**Author(s)**

Pattawee Puangchit

**See Also**

[get\\_plot\\_style\\_config](#), [get\\_export\\_config](#), [comparison\\_plot](#), [detail\\_plot](#), [stack\\_plot](#)

**Examples**

```
# Get all plot configurations with default settings
all_configs <- get_all_config()

# Get only comparison plot configuration
comp_config <- get_all_config("default")
```

---

get\_color\_palette

*Print and Visualize Themed Color Palettes*

---

**Description**

This function prints and visualizes predefined color palettes. Users can specify a 'color\_tone' and 'palette\_type' to display the corresponding colors. Additionally, calling 'color\_tone = "all"' returns a list of callable functions, where each entry dynamically generates a color palette visualization.

## Usage

```
get_color_palette(color_tone = NULL, palette_type = "qualitative")
```

## Arguments

- |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| color_tone   | <p>Character. The name of the predefined color theme. Default is 'NULL', which requires specification. Available themes include:</p> <ul style="list-style-type: none"> <li>• <b>Academic:</b> A balanced set of colors for research-oriented visuals.</li> <li>• <b>Purdue:</b> Themed after Purdue University branding.</li> <li>• <b>Colorblind:</b> Designed for colorblind-friendly visualization.</li> <li>• <b>Economic:</b> Used for economic data visualization.</li> <li>• <b>Trade:</b> Suited for trade-related data.</li> <li>• <b>GTAP:</b> Based on Global Trade Analysis Project visuals.</li> <li>• <b>GTAP2:</b> An alternative GTAP color set.</li> <li>• <b>Earth:</b> Inspired by natural, earthy tones.</li> <li>• <b>Vibrant:</b> High-contrast and energetic colors.</li> <li>• <b>Bright:</b> Bright and playful color combinations.</li> <li>• <b>Minimal:</b> Monochrome and muted colors for minimalistic designs.</li> <li>• <b>Energetic:</b> Warm and dynamic tones.</li> <li>• <b>Pastel:</b> Soft pastel shades.</li> <li>• <b>Spring:</b> Fresh, floral-inspired hues.</li> <li>• <b>Summer:</b> Sunny, warm color gradients.</li> <li>• <b>Winter:</b> Cool, icy tones for winter visuals.</li> <li>• <b>Fall:</b> Autumn-inspired warm color palette.</li> <li>• <b>Blue_mono, Green_mono, Red_mono, Grey_mono:</b> Monochromatic shades for respective colors.</li> </ul> <p>Use "all" to return a list of callable functions for all palettes.</p> |
| palette_type | <p>Character. The type of color palette to use. Options include:</p> <ul style="list-style-type: none"> <li>• <b>"qualitative"</b> - Best for categorical data.</li> <li>• <b>"sequential"</b> - Used for ordered, continuous scales.</li> <li>• <b>"diverging"</b> - Ideal for highlighting contrasts.</li> </ul> <p>Default is "qualitative".</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Details

The function retrieves colors from `create_color_palette()` and provides a visual preview of the selected theme.

If `color_tone = "all"`, the function returns a **list of functions**, where calling any element (e.g., `all_palettes$winter()`) generates the corresponding color palette visualization.

If the requested `color_tone` does not exist or is empty, the function throws an error.

## Value

- If a specific `color_tone` is provided, it prints the color palette and returns 'NULL'.
- If `color_tone = "all"`, it returns a **list of callable functions** to visualize each palette on demand.

**Author(s)**

Pattawee Puangchit

**Examples**

```
## Not run:
# Print & visualize a specific palette
get_color_palette("winter")

# Print & visualize another palette with different types
get_color_palette("fall", "sequential")
get_color_palette("academic", "diverging")

# Get all palettes as a list of callable functions
all_palettes <- get_color_palette("all")

# Click or call a specific palette function to view its colors
all_palettes$winter() # View the winter palette
all_palettes$fall()   # View the fall palette
all_palettes$gtap()   # View the GTAP palette

## End(Not run)
```

---

get\_export\_config

*Get Export Configuration Options*


---

**Description**

Returns documentation and default values for export configuration options used in plotting functions.

**Usage**

```
get_export_config(as_dataframe = TRUE, printing = FALSE)
```

**Arguments**

|              |                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------|
| as_dataframe | Logical. Whether to return settings as a dataframe. Default is FALSE.                                                            |
| printing     | Logical. If TRUE, prints a formatted code snippet that can be copied and pasted to recreate the configuration. Default is FALSE. |

**Details**

```
## **Export Configuration Parameters** - 'file_name': Character. Base name for exported files.
Default: "gtap_plots" - 'width': Numeric or 'NULL'. Plot width in inches. Default: 'NULL'
(automatically calculated) - 'height': Numeric or 'NULL'. Plot height in inches. Default: 'NULL'
(automatically calculated) - 'dpi': Numeric. Resolution for PNG export. Default: '300' - 'bg':
Character. Background color. Default: "white" - 'limitsize': Logical. Whether to limit size.
Default: 'FALSE'
```

**Value**

If `as_dataframe` is `TRUE`, returns a dataframe with export configuration settings. Otherwise, returns a list with configuration settings.

**Author(s)**

Pattawee Puangchit

**See Also**

[comparison\\_plot](#), [detail\\_plot](#), [stack\\_plot](#)

**Examples**

```
# Get export configuration as list
export_info <- get_export_config()

# Get as a formatted dataframe
export_df <- get_export_config(as_dataframe = TRUE)
```

---

`get_plot_style_config` *Get Plot Style Configuration*

---

**Description**

Returns configuration settings for plot styles, with options to view as a structured dataframe or to look up specific parameters. Also provides parameter validation for custom configurations.

**Usage**

```
get_plot_style_config(
  plot_type = "default",
  parameter_name = NULL,
  show_docs = FALSE,
  validate_custom = NULL,
  as_dataframe = FALSE,
  printing = TRUE
)
```

**Arguments**

|                              |                                                                                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>plot_type</code>       | Character. Type of plot: "default" (default).                                                                                                                |
| <code>parameter_name</code>  | Character or NULL. Name of specific parameter to return information about.                                                                                   |
| <code>show_docs</code>       | Logical. Whether to include documentation in the output.                                                                                                     |
| <code>validate_custom</code> | List or NULL. Custom configuration settings to validate.                                                                                                     |
| <code>as_dataframe</code>    | Logical. Whether to return settings as a dataframe.                                                                                                          |
| <code>printing</code>        | Logical. If <code>TRUE</code> , prints a formatted code snippet that can be copied and pasted to recreate the configuration. Default is <code>FALSE</code> . |

## Details

This function applies default styling for different types of plots and allows users to customize the appearance. The parameters are grouped as follows:

**## \*\*Title Settings\*\*** - 'show\_title': Logical. Show or hide the plot title. Default: 'TRUE' - 'title\_face': Character. Font face ("bold", "plain", "italic"). Default: "bold" - 'title\_size': Numeric. Font size of title. Default: '20' - 'title\_hjust': Numeric. Horizontal alignment (0 = left, 1 = right). Default: '0.5' - 'add\_unit\_to\_title': Logical. Append unit to title if applicable. Default: 'TRUE' - 'title\_margin': Numeric vector 'c(top, right, bottom, left)'. Default: 'c(10, 0, 10, 0)' - 'title\_format': List or NULL. Formatting options for the title, with elements:

- type: Character. One of "prefix", "suffix", "full", or "dynamic".
- text: Character. Text to add, or column names used for dynamic titles.
- sep: Character. Separator used for dynamic titles. Default: " - ".

**## \*\*X-Axis Settings\*\*** - 'show\_x\_axis\_title': Logical. Show or hide x-axis title. Default: 'TRUE' - 'x\_axis\_title\_face': Character. Font face for x-axis title. Default: "bold" - 'x\_axis\_title\_size': Numeric. Font size of x-axis title. Default: '16' - 'x\_axis\_title\_margin': Numeric vector 'c(top, right, bottom, left)'. Default: 'c(25, 25, 0, 0)' - 'show\_x\_axis\_labels': Logical. Show or hide x-axis labels. Default: 'TRUE' - 'x\_axis\_text\_face': Character. Font face for x-axis labels. Default: "bold" - 'x\_axis\_text\_size': Numeric. Font size of x-axis labels. Default: '14' - 'x\_axis\_text\_angle': Numeric. Angle of x-axis labels. Default: '45' - 'x\_axis\_text\_hjust': Numeric. Horizontal justification of x-axis labels. Default: '1' - 'x\_axis\_description': Character. Optional description for the x-axis. Default: ""

**## \*\*Y-Axis Settings\*\*** - 'show\_y\_axis\_title': Logical. Show or hide y-axis title. Default: 'TRUE' - 'y\_axis\_title\_face': Character. Font face for y-axis title. Default: "bold" - 'y\_axis\_title\_size': Numeric. Font size of y-axis title. Default: '16' - 'y\_axis\_title\_margin': Numeric vector 'c(top, right, bottom, left)'. Default: 'c(25, 25, 0, 0)' - 'show\_y\_axis\_labels': Logical. Show or hide y-axis labels. Default: 'TRUE' - 'y\_axis\_text\_face': Character. Font face for y-axis labels. Default: "plain" - 'y\_axis\_text\_size': Numeric. Font size of y-axis labels. Default: '14' - 'y\_axis\_text\_angle': Numeric. Angle of y-axis labels. Default: '0' - 'y\_axis\_text\_hjust': Numeric. Horizontal justification of y-axis labels. Default: '0' - 'y\_axis\_description': Character. Optional description for the y-axis. Default: "" - 'show\_axis\_titles\_on\_all\_facets': Logical. Show axis titles on all facets. Default: 'TRUE'

**## \*\*Value Label Settings\*\*** - 'show\_value\_labels': Logical. Show or hide value labels. Default: 'TRUE' - 'value\_label\_face': Character. Font face for value labels. Default: "plain" - 'value\_label\_size': Numeric. Font size of value labels. Default: '5' - 'value\_label\_position': Character. Position of value labels ("above", "outside", "top"). Default: "above" - 'value\_label\_decimal\_places': Numeric. Number of decimal places in value labels. Default: '2'

**## \*\*Legend Settings\*\*** - 'show\_legend': Logical. Show or hide legend. Default: 'FALSE' - 'show\_legend\_title': Logical. Show or hide legend title. Default: 'FALSE' - 'legend\_position': Character. Legend position ("none", "bottom", "right"). Default: "none" - 'legend\_title\_face': Character. Font face for legend title. Default: "bold" - 'legend\_text\_face': Character. Font face for legend text. Default: "plain" - 'legend\_text\_size': Numeric. Font size for legend text. Default: '14'

**## \*\*Panel Strip Settings\*\*** - 'strip\_face': Character. Font face for panel strip. Default: "bold" - 'strip\_text\_size': Numeric. Font size for panel strip. Default: '16' - 'strip\_background': Character. Background color of strip. Default: "lightgrey" - 'strip\_text\_margin': Numeric vector 'c(top, right, bottom, left)'. Default: 'c(10, 0, 10, 0)'

**## \*\*Panel Layout\*\*** - 'panel\_spacing': Numeric. Spacing between panels. Default: '2' - 'panel\_rows': Numeric or 'NULL'. Number of rows in panel layout. Default: 'NULL' - 'panel\_cols': Numeric



or 'NULL'. Number of columns in panel layout. Default: 'NULL' - 'theme': ggplot2 theme object or 'NULL'. Custom ggplot theme. Default: 'NULL'

## \*\*Color and Grid Settings\*\* - 'color\_tone': Character or 'NULL'. Base color theme. Default: 'NULL' - 'positive\_color': Character. Color for positive values. Default: "#2E8B57" - 'negative\_color': Character. Color for negative values. Default: "#CD5C5C" - 'background\_color': Character. Background color of plot. Default: "white" - 'grid\_color': Character. Color of grid lines. Default: "grey90" - 'show\_grid\_major\_x': Logical. Show major grid lines on x-axis. Default: 'FALSE' - 'show\_grid\_major\_y': Logical. Show major grid lines on y-axis. Default: 'TRUE' - 'show\_grid\_minor\_x': Logical. Show minor grid lines on x-axis. Default: 'FALSE' - 'show\_grid\_minor\_y': Logical. Show minor grid lines on y-axis. Default: 'FALSE'

## \*\*Zero Line Settings\*\* - 'show\_zero\_line': Logical. Show or hide zero line. Default: 'TRUE' - 'zero\_line\_type': Character. Line type ("solid", "dashed", "dotted"). Default: "dashed" - 'zero\_line\_color': Character. Color of zero line. Default: "black" - 'zero\_line\_size': Numeric. Line thickness of zero line. Default: '0.5' - 'zero\_line\_position': Numeric. Position of the zero line. Default: '0'

## \*\*Bar Chart Settings\*\* - 'bar\_width': Numeric. Width of bars. Default: '0.9' - 'bar\_spacing': Numeric. Spacing between bars. Default: '0.9'

## \*\*Scale Settings\*\* - 'scale\_limit': Numeric vector of length 2. Manual limits for value axis. Example: 'c(-10, 10)' - 'scale\_increment': Numeric. Step size for axis tick marks. Example: '2'

## \*\*Scale Expansion Settings\*\* - 'expansion\_y\_mult': Numeric vector. Y-axis expansion. Default: 'c(0.05, 0.1)' - 'expansion\_x\_mult': Numeric vector. X-axis expansion. Default: 'c(0.05, 0.05)'

## \*\*All Font Adjustment\*\* - 'all\_font\_size': Numeric. Master control for all font sizes. Default: '1'

## \*\*Plot Margin Settings\*\* - 'plot.margin': Numeric vector 'c(top, right, bottom, left)'. Margins around the entire plot. Default: 'c(10, 25, 10, 10)'

## Value

If parameter\_name is specified, returns a list with the value and documentation for that parameter. If validate\_custom is specified, returns the validated configuration list. If as\_dataframe is TRUE, returns a dataframe with configuration settings. Otherwise, returns a list with all configuration settings.

## Author(s)

Pattawee Puangchit

## See Also

[comparison\\_plot](#), [detail\\_plot](#), [stack\\_plot](#)

## Examples

```
# Get all default configuration
get_plot_style_config(printing = TRUE)

# Get information about a specific parameter
param_info <- get_plot_style_config("default", "bar_width")

# Get as structured dataframe
config_df <- get_plot_style_config("default", as_dataframe = TRUE)
```

```
# Validate custom configuration
custom_config <- list(title_size = 24, bar_width = 0.7, plot.margin = c(10, 50, 10, 10))
validated <- get_plot_style_config("default", validate_custom = custom_config)
```

---

gtap\_macros\_data

*Extract and Aggregate Scalar Macroeconomic Variables*


---

## Description

Extracts scalar macroeconomic variables from multiple SL4 datasets and aggregates them into a structured data frame.

## Usage

```
gtap_macros_data(
  select_var = NULL,
  experiment = NULL,
  input_path = NULL,
  output_path = NULL,
  output_formats = NULL,
  subtotal_level = FALSE
)
```

## Arguments

|                |                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------|
| select_var     | Character vector (optional). List of specific variable names to filter from the final result. If NULL, all variables are returned. |
| experiment     | Character vector. List of experiment names corresponding to SL4 files.                                                             |
| input_path     | Character. Path to the directory containing SL4 files.                                                                             |
| output_path    | Character (optional). Directory to save exported data.                                                                             |
| output_formats | Character vector (optional). List of output formats (e.g., "csv", "xlsx").                                                         |
| subtotal_level | Logical. Whether to include subtotal levels in the processed data.                                                                 |

## Value

A sorted data frame containing processed GTAP macro data.

## Author(s)

Pattawee Puangchit

## See Also

[add\\_mapping\\_info](#), [auto\\_gtap\\_data](#)

## Examples

```
# Input Path:
input_path <- system.file("extdata/in", package = "GTAPViz")

# GTAP Macro Variables from 2 .sl4 Files named (EXP1, EXP2)
# Note: No need to add .sl4 to the experiment name
gtap_macro <- gtap_macros_data(NULL, experiment = c("EXP1", "EXP2"),
                               input_path = input_path, subtotal_level = FALSE)
```

---

`pivot_table_with_filter`

*Export Data as an Excel Pivot Table*

---

## Description

Exports a dataset to an Excel file with both raw data and a generated pivot table.

## Usage

```
pivot_table_with_filter(
  data,
  filter = NULL,
  rows = NULL,
  cols = NULL,
  data_fields = "Value",
  raw_sheet_name = "RawData",
  pivot_sheet_name = "PivotTable",
  dims = "A5",
  export = TRUE,
  output_path = getwd(),
  workbook_name = "GTAP_PivotTable.xlsx"
)
```

## Arguments

|                               |                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------|
| <code>data</code>             | Data frame. The dataset to be exported.                                                   |
| <code>filter</code>           | Character vector (optional). Columns to be used as filter fields in the pivot table.      |
| <code>rows</code>             | Character vector (optional). Columns to be used as row fields in the pivot table.         |
| <code>cols</code>             | Character vector (optional). Columns to be used as column fields in the pivot table.      |
| <code>data_fields</code>      | Character. The data field(s) to be summarized in the pivot table (default: "Value").      |
| <code>raw_sheet_name</code>   | Character. Name of the sheet containing raw data (default: "RawData").                    |
| <code>pivot_sheet_name</code> | Character. Name of the sheet containing the pivot table (default: "PivotTable").          |
| <code>dims</code>             | Character. Cell reference where the pivot table starts (default: "A3").                   |
| <code>export</code>           | Logical. Whether to save the Excel file (default: TRUE).                                  |
| <code>output_path</code>      | Character. Directory where the file should be saved (default: current working directory). |
| <code>workbook_name</code>    | Character. Name of the output Excel file (default: "GTAP_PivotTable.xlsx").               |

**Details**

This function creates an Excel workbook with: - A raw data sheet ('raw\_sheet\_name') containing the provided dataset. - A pivot table sheet ('pivot\_sheet\_name') generated based on specified row, column, and data fields.

If 'export = TRUE', the function saves the workbook to the specified 'output\_path'.

**Value**

An excel workbook object containing both raw data and the pivot table.

**Author(s)**

Pattawee Puangchit

**Examples**

```
# Input Path
input_path <- system.file("extdata/in", package = "GTAPViz")

# Prepares data from SL4 files for EXP1 and EXP2
auto_gtap_data(
  experiment = c("EXP1", "EXP2"),
  input_path = input_path,
  subtotal_level = FALSE,
  process_sl4_vars = c("qo", "qgdp"),
  process_har_vars = FALSE,
  mapping_info = "GTAPv7",
  plot_data = TRUE
)

# Generate comparison table
pivot_table_with_filter(
  data = sl4.plot.data[["2D"]][["Sector"]],
  filter = c("Variable", "Unit"),
  rows = c("Region", "Sector"),
  cols = c("Experiment"),
  data_fields = "Value",
  raw_sheet_name = "Raw_Data",
  pivot_sheet_name = "Sector_Pivot",
  export = TRUE,
  output_path = tempdir(),
  workbook_name = "Sectoral_Impact_Analysis.xlsx"
)
```

---

rename\_GTAP\_bilateral *Rename GTAP Bilateral Trade Columns*

---

**Description**

Renames bilateral trade columns in GTAP data to standardized names, ensuring consistency in regional trade flows.

**Usage**

```
rename_GTAP_bilateral(data)
```

**Arguments**

`data` Data structure containing GTAP bilateral trade data.

**Value**

The same data structure with renamed bilateral trade columns.

**Author(s)**

Pattawee Puangchit

**See Also**

[add\\_mapping\\_info](#), [convert\\_units](#), [sort\\_plot\\_data](#)

**Examples**

```
# Load Sample Data:
sl4_data1 <- HARplus::load_sl4x(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPViz"))

# Get Data by Variable Name
sl4_data1 <- HARplus::get_data_by_var("qxs", sl4_data1)

# Rename bilateral trade columns in a GTAP dataset
gtap_data <- rename_GTAP_bilateral(sl4_data1)
```

---

|              |                                  |
|--------------|----------------------------------|
| rename_value | <i>Rename Values in a Column</i> |
|--------------|----------------------------------|

---

**Description**

Replaces specific values in a column based on a provided mapping file. Supports renaming across nested data structures and preserves factor levels.

**Usage**

```
rename_value(data, column_name = NULL, mapping.file)
```

**Arguments**

`data` Data structure (data frame, list, or nested combination).  
`column_name` Character. Column to modify. If 'NULL', the function extracts it from 'mapping.file'.  
`mapping.file` Data frame with "OldName" and "NewName" columns for renaming.

**Value**

The same data structure with specified values replaced.

**Author(s)**

Pattawee Puangchit

**See Also**[add\\_mapping\\_info](#), [convert\\_units](#), [sort\\_plot\\_data](#)**Examples**

```
# Load Sample Data:
sl4_data1 <- HARplus::load_sl4x(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPViz"))

# Get Data by Variable Name
sl4_data1 <- HARplus::get_data_by_var(c("qgdp", "EV"),sl4_data1)

# Rename variables in a dataset
rename_map <- data.frame(OldName = c("qgdp", "EV"),
                          NewName = c("Real GDP", "Welfare"))
gtap_data <- rename_value(sl4_data1, column_name = "Variable",
                          mapping.file = rename_map)
```

report\_table

*Generate a Structured Report Table***Description**

Transforms multiple datasets into wide-format tables based on defined pivot columns, hierarchical grouping, and renaming rules. Supports optional subtotal filtering and exporting to Excel.

**Usage**

```
report_table(
  data_list,
  pivot_col,
  total_column = FALSE,
  export_table = FALSE,
  separate_file = FALSE,
  output_path = NULL,
  sheet_names = NULL,
  include_units = FALSE,
  component_exclude = NULL,
  group_by = NULL,
  rename_cols = NULL,
  var_name_by_description = TRUE,
  add_var_info = FALSE,
  decimal = 2,
  unit_select = NULL,
  separate_sheet_by = NULL,
  subtotal_level = FALSE,
  repeat_label = FALSE,
  workbook_name = "detail_results",
```

```

    add_group_line = FALSE
  )

```

### Arguments

|                         |                                                                                                                                                                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data_list               | A named list of data frames to process.                                                                                                                                                                                               |
| pivot_col               | A named list specifying the column to pivot into a wide format for each dataset. Each dataset can have only one pivot column. Example: <code>pivot_col = list(A = "COLUMN", E1 = "PRICES")</code>                                     |
| total_column            | Logical. If 'TRUE', adds a "Total" column summing numeric values.                                                                                                                                                                     |
| export_table            | Logical. If 'TRUE', saves the output as an Excel file.                                                                                                                                                                                |
| separate_file           | Logical. If 'TRUE', saves each dataset as a separate Excel file.                                                                                                                                                                      |
| output_path             | Character. Directory for saving Excel files when 'export_table = TRUE'.                                                                                                                                                               |
| sheet_names             | Optional named list for custom sheet names.                                                                                                                                                                                           |
| include_units           | Logical. If 'TRUE', includes "Unit" as a grouping column if applicable.                                                                                                                                                               |
| component_exclude       | Optional character vector specifying pivoted values to exclude.                                                                                                                                                                       |
| group_by                | A named list defining hierarchical grouping for each dataset. The order of columns in each list determines the priority. Example: <code>group_by = list(A = list("Experiment", "REG"), E1 = list("Experiment", "REG", "COMM"))</code> |
| rename_cols             | A named list for renaming columns across <b>all</b> datasets. Example: <code>rename_cols = list("REG" = "Region", "COMM" = "Commodities", "Experiment" = "Scenario")</code>                                                           |
| var_name_by_description | Logical. If 'TRUE', replaces variable codes with descriptions when available.                                                                                                                                                         |
| add_var_info            | Logical. If 'TRUE', appends variable codes in parentheses after descriptions.                                                                                                                                                         |
| decimal                 | Numeric. Number of decimal places for rounding values.                                                                                                                                                                                |
| unit_select             | Optional character. Specifies a unit to filter the dataset.                                                                                                                                                                           |
| separate_sheet_by       | Optional column name to split sheets in Excel. If defined, each unique value in the specified column gets its own sheet. Example: <code>separate_sheet_by = "Scenario"</code> .                                                       |
| subtotal_level          | Logical. If 'TRUE', includes all subtotal values; otherwise, keeps only 'TOTAL' rows.                                                                                                                                                 |
| repeat_label            | Logical. If 'TRUE', repeats the first group column in exports for clarity.                                                                                                                                                            |
| workbook_name           | Character. Name of the Excel workbook (without extension).                                                                                                                                                                            |
| add_group_line          | Logical. If 'TRUE', adds a thin line after each group in the exported table.                                                                                                                                                          |

### Value

A named list of transformed data frames. If 'export\_table = TRUE', tables are saved as Excel files.

### Author(s)

Pattawee Puangchit

### See Also

[add\\_mapping\\_info](#), [convert\\_units](#), [rename\\_value](#)

## Examples

```
# Input Path
input_path <- system.file("extdata/in", package = "GTAPViz")

# Prepares data from SL4 files for EXP1 and EXP2
auto_gtap_data(
  experiment = c("EXP1", "EXP2"),
  input_path = input_path,
  subtotal_level = FALSE,
  process_sl4_vars = c("qgdp", "EV", "qo"),
  process_har_vars = FALSE,
  mapping_info = "GTAPv7",
  plot_data = TRUE
)

# Generate comparison table
report_table(
  data_list = sl4.plot.data[["1D"]],
  pivot_col = list(Region = "Variable"),
  group_by = list(Region = list("Experiment", "Region")),
  rename_cols = list("Experiment" = "Scenario"),

  total_column = FALSE,
  decimal = 4,
  subtotal_level = FALSE,
  repeat_label = FALSE,
  include_units = TRUE,

  var_name_by_description = TRUE,
  add_var_info = TRUE,
  add_group_line = FALSE,

  separate_sheet_by = "Unit",
  export_table = FALSE,
  output_path = tempdir(),
  separate_file = FALSE,
  workbook_name = "Comparison Table"
)
```

---

sort\_plot\_data

---

Sort GTAP Plot Data

---

## Description

Sorts data for plotting using flexible options for column ordering and value-based sorting. Works with data frames, lists of data frames, or nested data structures.

## Usage

```
sort_plot_data(
  data,
  sort_columns = NULL,
  sort_by_value_desc = NULL,
```





stack\_plot

*Create Stacked Bar Charts for Decomposition Analysis***Description**

Generates stacked bar charts to visualize value compositions across multiple dimensions. The function supports stacked and unstacked presentations for decomposition analysis.

**Usage**

```
stack_plot(
  data,
  filter_var = NULL,
  x_axis_from,
  stack_value_from,
  split_by = NULL,
  panel_var = "Experiment",
  variable_col = "Variable",
  unit_col = "Unit",
  desc_col = "Description",
  var_name_by_description = FALSE,
  add_var_info = FALSE,
  show_total = TRUE,
  unstack_plot = FALSE,
  top_impact = NULL,
  invert_pane = FALSE,
  separate_figure = FALSE,
  output_path = NULL,
  export_picture = TRUE,
  export_as_pdf = FALSE,
  export_config = NULL,
  plot_style_config = NULL
)
```

**Arguments**

|                  |                                                                                                                                                                                                      |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data             | A data frame or a list of data frames containing GTAP results.                                                                                                                                       |
| filter_var       | Vector or data frame. If a vector, filters the values in 'x_axis_from'. If a data frame, filters 'value_col' based on matching 'variable_col' values.                                                |
| x_axis_from      | Character. Column name for x-axis categories (e.g., "REG", "Sector").                                                                                                                                |
| stack_value_from | Character. Column containing stack component categories (e.g., "COMM" for commodities).                                                                                                              |
| split_by         | Character or vector. Column name(s) to generate separate plots for each unique value (e.g., "COMM", "REG", "Variable"). NULL creates a single aggregated plot, appropriate for macro-level analysis. |
| panel_var        | Character. Column for panel facets (default: "Experiment").                                                                                                                                          |
| variable_col     | Character. Column containing variable identifiers (default: "Variable").                                                                                                                             |
| unit_col         | Character. Column containing unit information (default: "Unit").                                                                                                                                     |

|                         |                                                                                                                                                                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| desc_col                | Character. Column containing variable descriptions (default: "Description").                                                                                                                                                                |
| var_name_by_description | Logical. If TRUE, uses descriptions instead of variable codes in titles (default: FALSE).                                                                                                                                                   |
| add_var_info            | Logical. If TRUE, adds the variable code in parentheses after the description (default: FALSE).                                                                                                                                             |
| show_total              | Logical. If TRUE, displays total values above stacked bars (default: TRUE).                                                                                                                                                                 |
| unstack_plot            | Logical. If TRUE, creates separate bar plots for each 'x_axis_from' value instead of stacked bars (default: FALSE).                                                                                                                         |
| top_impact              | Numeric or NULL. If specified, shows only the top N impactful values; NULL shows all values.                                                                                                                                                |
| invert_pane             | Logical. If TRUE, creates horizontal bars instead of vertical ones (default: FALSE).                                                                                                                                                        |
| separate_figure         | Logical. If TRUE, generates separate figures per panel value (default: FALSE).                                                                                                                                                              |
| output_path             | Character. Directory path for saving output files. If NULL, plots are only returned in R.                                                                                                                                                   |
| export_picture          | Logical. If TRUE, exports plots as image files (default: TRUE).                                                                                                                                                                             |
| export_as_pdf           | Logical or "merged". If TRUE, exports separate PDFs; if "merged", creates a multi-page PDF; if FALSE, skips PDF (default: FALSE).                                                                                                           |
| export_config           | List. Export settings, including: <ul style="list-style-type: none"> <li>• 'width': Output file width (in inches).</li> <li>• 'height': Output file height (in inches).</li> <li>• Additional settings—see '?get_export_config'.</li> </ul> |
| plot_style_config       | List. Custom plot styles—see '?get_plot_style_config'.                                                                                                                                                                                      |

**Value**

A ggplot2 object for a single plot or a list of ggplot2 objects for multiple plots.

**Author(s)**

Pattawee Puangchit

**See Also**

[get\\_plot\\_style\\_config](#), [get\\_export\\_config](#), [comparison\\_plot](#), [detail\\_plot](#)

**Examples**

```
# Input Path:
input_path <- system.file("extdata/in", package = "GTAPViz")

# GTAP Macro Variables from 2 .sl4 Files named (EXP1, EXP2)
# Note: No need to add .sl4 to the experiment name
gtap_data <- auto_gtap_data(experiment = c("EXP1", "EXP2"),
                           input_path = input_path, subtotal_level = FALSE,
                           process_sl4_vars = NULL, process_har_vars = NULL,
                           mapping_info = "GTAPv7", plot_data = TRUE)
```

```
stack_plot(data = har.plot.data[["A"]],
           x_axis_from = "REG",
           stack_value_from = "COLUMN",
           split_by = FALSE,

           show_total = TRUE,
           unstack_plot = FALSE,

           var_name_by_description = TRUE,

           invert_pane = FALSE,
           separate_figure = FALSE,

           export_picture = FALSE,
           export_as_pdf = FALSE,
           export_config = list(
             width = 28,
             height = 15
           ),
           output_path = tempdir(),

           plot_style_config = list(
             color_tone = "gtap",
             panel_rows = 2,
             panel_cols = NULL,
             show_legend = TRUE,
             show_axis_titles_on_all_facets = FALSE
           ))
```

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